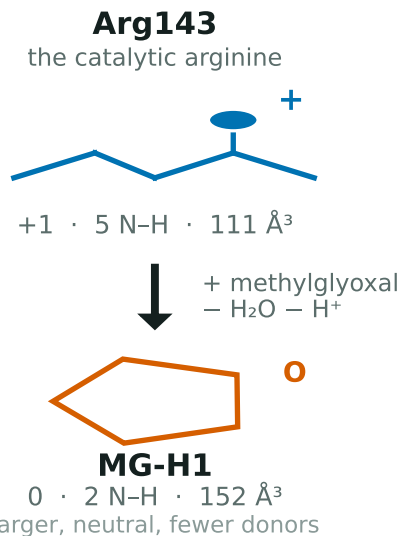
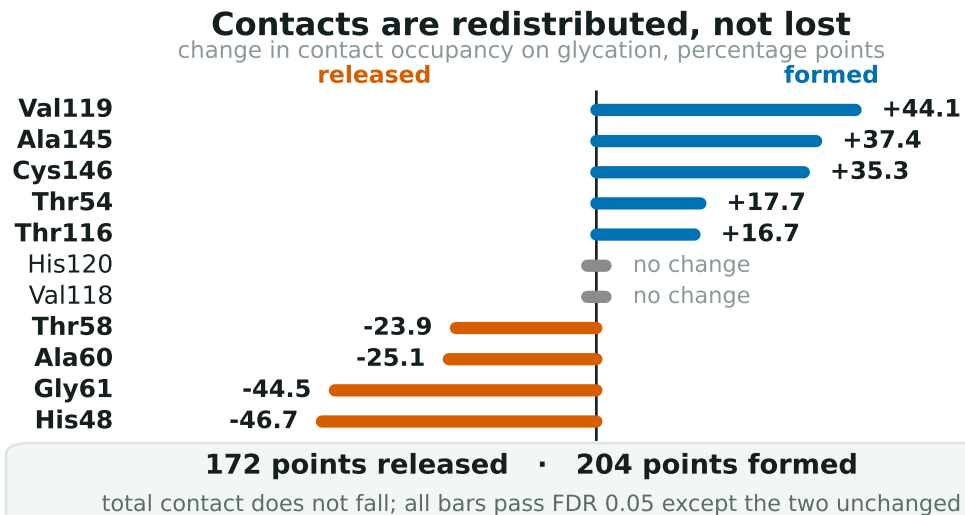


## 1 · THE MODIFICATION



## 2 · WHAT GLYCATION DOES TO THE SITE



## 3 · THE PRIMARY ENDPOINT

**Site-143 composite**  
His48 / Cys57 / Gly61

**-36.1**

percentage points

**p = 0.0009 vs  $\alpha$  = 0.0042**  
n = 10 per group · explicit OPC water

**His48 and Gly61 move together**  
r = +0.995 across replicates;  
Cys57 varies independently

RMSD, R<sub>g</sub>,  $\beta$ -sheet:  
**no detectable change**

### One-residue difference

identical backbone coordinates;  
the adduct is the only variable

### Replicate-level

n = 10 per group; the replicate  
is the unit, never the frame

### Robust

holds across five analysis windows  
and four contact criteria

### Fully reported

negative and inconclusive  
comparisons reported alongside

## Glycation at Arg143 alters local loop-junction contacts without detectable global destabilisation

The enlarged, charge-neutralised side chain releases its contacts to the zinc-binding loop and forms others toward Val119, Ala145 and Cys146. Backbone RMSD, radius of gyration and  $\beta$ -sheet content show no detectable change across twenty replicates: the perturbation is local, not global.